

CORE THESIS

Regulated workflows are fragmented. Teams are forced to carry quality context between systems. PixeSci closes the QC execution gap by connecting existing instruments, software, procedures and quality systems - executing approved workflows while preserving context, approvals and audit-ready evidence.

1. PROBLEM

QC in regulated life sciences is fragmented across multiple systems, teams and external partners that do not naturally operate together, creating manual integrations that lead to time inefficiencies, loss of context and highly expensive errors.

QC work spans instruments, CDS, LIMS, QMS, environmental monitoring, equipment, procedures, documents and external partners that do not naturally operate together.

Teams become the integration layer: moving data, reconstructing context, checking conditions, coordinating handoffs, initiating exceptions, obtaining approvals and proving what happened.

The deeper problem is a **fragmented compliance evidence chain** that is difficult to continuously execute, monitor and review.

2. CONSEQUENCES

Small evidence gaps can become expensive reconstruction problems.

Operational: rework, investigations, deviations, delayed review/release, batch holds.

Regulatory: retrospective review, CAPA/remediation, warning letters, import restrictions.

Commercial: product loss, recalls, delayed market access and revenue risk.

Patient: when the underlying failure affects product quality, risk can ultimately reach patients.

3. ALTERNATIVES

- Manual review / data entry
- Spreadsheets / email / shared drives
- Paper / e-forms
- LIMS / QMS / CDS / ELN
- Point-to-point integrations
- Vendor automation
- Scripts / APIs
- RPA
- Consultants / integrators
- Additional QC / QA staff

Limitation: These solve portions of the process, but regulated workflows still cross their boundaries. Teams remain responsible for carrying context across the gaps.

8. WHY NOW

The race to implement AI is underway.

Life-science companies are moving from asking "Can we use AI?" to "How do we actually implement it?"

But regulated operations cannot simply adopt AI like other industries. AI must operate within validation, traceability, approvals, data integrity, auditability and Quality Unit oversight.

The opportunity is not just bringing AI into the lab. It is making AI operational in a regulated lab.

4. CUSTOMERS

Beachhead

- Emerging regulated life-science organizations with QC/manufacturing operations

Primary

- SMB biomanufacturers
- Biologics / sterile-product companies
- Quality-heavy biotech startups
- CDMOs / CMOs
- Regulated testing organizations

Under validation

- Emerging medical-device / IVD manufacturers
- Contract manufacturers

5. FILTERS

MUST HAVE

- 2–50 employees
- Regulated (GMP, 21 CFR Part 11, FDA QMSR, ISO 13485, ISO/IEC 17025, EU Annex 11)
- AI-receptive

GOOD TO HAVE

- Multi-system
- Documentation-heavy
- GMP transition
- Investigation-heavy
- Limited bandwidth
- Looking for AI deployment

6. VALUE PROP

We let quality teams in regulated life sciences talk to their lab.

PixeSci connects the lab - instruments, software, procedures and quality systems - so teams can ask what is happening, tell it what needs to happen, and coordinate permitted work while preserving compliance context, approvals and audit-ready evidence.

7. SOLUTION

Quality Core

Integration Layer - adapters/connectors.

Agentic Orchestration - permitted execution across systems.

Continuous Controls - workflow state and exception monitoring.

Evidence Layer - source linkage, lineage, actions and exceptions.

Deployment - **PixeSci Node**, PixeSci-shipped hardware installed at the customer site for local operation. **Instant deployment.**

9. CHANNELS

Founder-led discovery and sales; LinkedIn and targeted email; warm introductions; incubators and regional life-science ecosystems; conferences/events; design partners and pilots; QC/regulatory educational content; consultants/advisors; integration partners; referrals.

10. REVENUE

- **Startup** — **\$18K/year** | 10 users + \$3K / additional user / year
- **Tier 1** — **\$50K/year** | 25 users + \$4K / additional user / year
- **Tier 2** — **\$120K/year** | 50 users + \$5K / additional user / year
- **Enterprise** — **\$300K/year** | 100 users 100+ users: Contact Us
- **Integrations** — 5 scientific software / instruments included in Startup; **\$3K per additional integration**
- **Expansion** — additional users, integrations, sites, workflows, deployment and support

11. COSTS

PRODUCTIZED TARGET / CUSTOMER

- PixeSci Node / local compute — **\$1,400**
- Shipping — **\$100**
- Ready-to-use deployment — **\$150** (~5 hrs x \$30/hr)
- Ongoing support — **\$720/year** (~2 hrs/mo x 12 x \$30/hr)

- **Direct COGS** — **\$2,370/customer**
- **Base CAC** — **\$2,000/customer**

12. KEY METRICS

LOIs

Pipeline

Pilots

13. ADVANTAGES

Orchestration across modern and legacy lab environments; compliance context carried across systems; controlled agent execution; **PixeSci Node** local hardware; growing instrument/software adapter library; accumulated regulated-workflow knowledge; direct pharmaceutical/laboratory experience.

These are **developing advantages** - **not yet established moats.**

Manual transcription / spreadsheet calculations	Tismor: FDA described an incorrect assay result caused by a transcription entry error into an Excel spreadsheet used for batch-release calculations; product had already been released.	Controlled transfer, calculation provenance and immutable source linkage.
Instrument activity not reconciled to final result	Unipack: FDA described injections/runs that were not fully reconciled to the final reported record, including a stability sample reinjected after an earlier run had no reported result.	Run lineage, completeness checks and automatic routing of unusual activity for review.
Fragmented / missing raw records	FDA findings have described laboratory evidence split across electronic and paper records, missing data, and quality review that did not reliably reach underlying raw evidence.	Instrument-to-record linkage, evidence completeness and cross-system context.
Audit trails / electronic records not adequately reviewed	FDA letters repeatedly cite inadequate audit-trail review, weak electronic-record controls and insufficient Quality Unit oversight.	Continuous audit-trail ingestion, prioritized review, identity-aware activity evidence.
Environmental excursions not converted into timely investigations	Sterile-manufacturing findings include environmental-monitoring excursions/trends that were not investigated promptly.	Monitor -> detect exception -> initiate quality workflow -> route investigation.
Incomplete OOS / deviation investigations	FDA repeatedly requires retrospective assessment when OOS/deviation investigations are scientifically inadequate or fail to connect relevant evidence.	Cross-system case context, evidence packages and investigation completeness.
Quality-unit oversight / resource constraints	FDA findings can link data-review and transcription weaknesses to inadequate Quality Unit oversight or resources.	Automation should increase quality-team leverage without removing required human authorization.

FRAGMENTED QUALITY STACK	TEAMS CARRY CONTEXT	QC EXECUTION GAP	LATE DISCOVERY	CONSEQUENCES
Instruments • CDS • LIMS • QMS • SOPs • environmental systems	Move data • reconstruct context • check conditions • coordinate handoffs • prove execution	Approved process != reliably connected execution/evidence	Review • investigation • batch disposition • audit • inspection	Rework • delay • remediation • regulatory / commercial risk • potential product/patient risk

Source	Evidence / learning	Supports	Status
FDA warning-letter research	Recurring patterns: fragmented raw data, manual transcription, unreconciled runs, audit-trail review gaps, incomplete OOS/deviation investigations and retrospective lookbacks.	Problem severity + evidence-chain thesis	Strong external evidence
Voltas / Sep 18	Lean med-device manufacturer; 3-person quality/regulatory team; Grand Avenue eQMS; cleanroom, manufacturing, packaging and sterilization; limited bandwidth.	Medical-device segment + lean-team hypothesis	Strong signal; direct quality-user validation needed
Gerard / Sep 15	Deep LIMS workflow and implementation discussion covering samples, tests, environmental records, equipment, audit trails and e-signatures.	Orchestrate around systems of record	Strong domain evidence
Parser development / Sep 20	Format adapters -> unified schema; start narrow with HPLC/UPLC; validate missing/malformed data.	Concrete technical wedge	Strong product evidence
Existing pilots / discovery	Two early biotech pilots and broad discovery interest; quantified ROI and paid conversion not yet established.	Initial market pull	Early evidence
Pricing curve	\$18K -> \$50K -> \$120K -> \$300K annual tiers.	Commercial model to test	Hypothesis

Decision rule: broaden the beachhead only when repeated workflow evidence shows a painful problem, an identifiable buyer, measurable value and willingness to test or pay. PixeSci should not imply that every documentation gap causes patient harm; these are weaknesses in the control system designed to prevent quality failures from reaching patients.